and it is pointed out that, although investigations undertaken in various countries over recent years show encouraging results, the complexity of the problem makes progress slow. The areas of research considered of foremost importance in mental-health work are enumerated and discussed in detail. Among the many subjects recommended for investigation are studies on brain function, epidemiology, health and social studies of communities undergoing rapid change, the hospital *milieu*, ecology of mental illness, problems of ageing, the effect of nutrition on mental health, and genetics.

The report concludes with a consideration of the vital part that the World Health Organization can play in future planning for mental health, by fostering international co-operation and exchange of information, and by facilitating and stimulating research.

APPLICATION OF PALÆOMAGNETISM TO THRUST MECHANICS

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THE surface trace of the Laramide Lewis thrust fault in the south-eastern Cordillera of Canada exhibits a change in direction of approximately 35° from a north-west trend east of the Clark and Lewis Ranges to a north trend east of the Flathead Range (Fig. 1). The nature and genesis of this change of direction has been the subject of much controversy in recent years, and explanations have included differential erosion of the thrust plate and differential rotation of the plate. The purpose of this article is to show that palæomagnetic directions in conjunction with surface geological data suggest that the Lewis thrust plate moved as a coherent tectonic unit and that the gross geometry of the plate may have been established at the time thrusting was initiated.

For convenience in reference, the Lewis thrust plate will be divided geographically into two parts, termed respectively the Flathead Range segment and the Clark and Lewis Ranges segment.

These two parts are structurally and stratigraphically continuous with one another at their common boundary in the vicinity of North Kootenay Pass. Precambrian rocks of the Purcell system have been mapped without interruption between them, and no major transverse faults are there to suggest that the two segments of the Lewis thrust plate operated as independent tectonic units[1]. The continuity as well as the simplicity of the structure between the two segments would suggest rather that the plate moved as a single structural unit.

The change in trend of the surface trace of the Lewis thrust therefore must be due either to differential erosion of the plate, such that the Flathead Range segment has been eroded to some considerable degree farther west than the Clark and Lewis Ranges segment, or to the possibility that the latter segment was always structurally farther east than the former. The fact, however, that the fold and thrust fault elements in the Clark Range do not end abruptly at the margin of the plate where the surface trace of the Lewis thrust swings west but rather swing sub-parallel to it suggests that the re-entrant at North Kootenay Pass is not due to differential erosion of the Lewis thrust plate. The possibility of differential rotation, therefore, is especially impor-

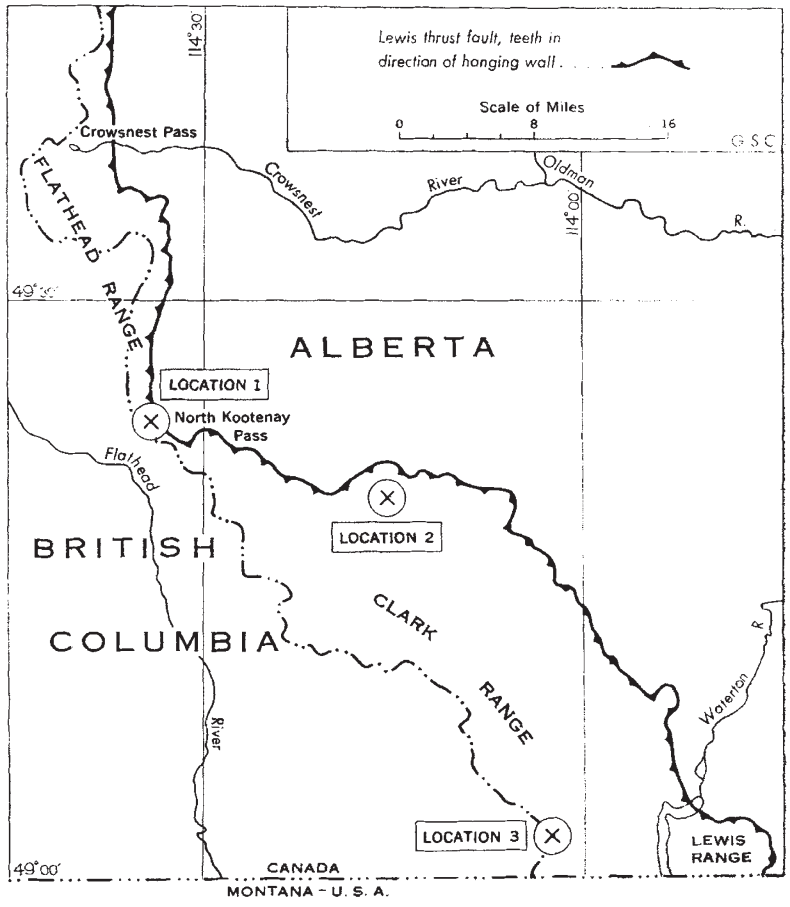


Fig. 1. Location map

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tant in the interpretation of the mechanics of emplacement of the plate.

If differential rotation has occurred it should be possible to determine the approximate amount of relative motion by a comparison of the directions of the remanent magnetic vectors obtained from Purcell samples collected in the two segments. A necessary condition, however, is that the time of magnetization of the rocks making up the plate be pre-Laramide (older than Early Tertiary) in order that the results obtained can be applied to this problem.

Fifty-nine samples of unweathered, highly quartzose, fine-grained, red sandstone were collected from the Precambrian Purcell system at intervals normal to the strike at three locations in the Lewis thrust plate (Fig. 1). They were from member *A* of the Kintla formation in the Flathead Range where the strata strike north, and, from members *A* and *C* of the same formation, at two localities in the Clark Range where the strata strike north-west.

The remanent magnetization in each of two cubes from each sample of Kintla formation was determined with a sensitive astatic magnetometer in the laboratories of the Dominion Observatory of Canada. From the measurements of two cubes the mean attitude of the remanent magnetic moment was computed for each sample.

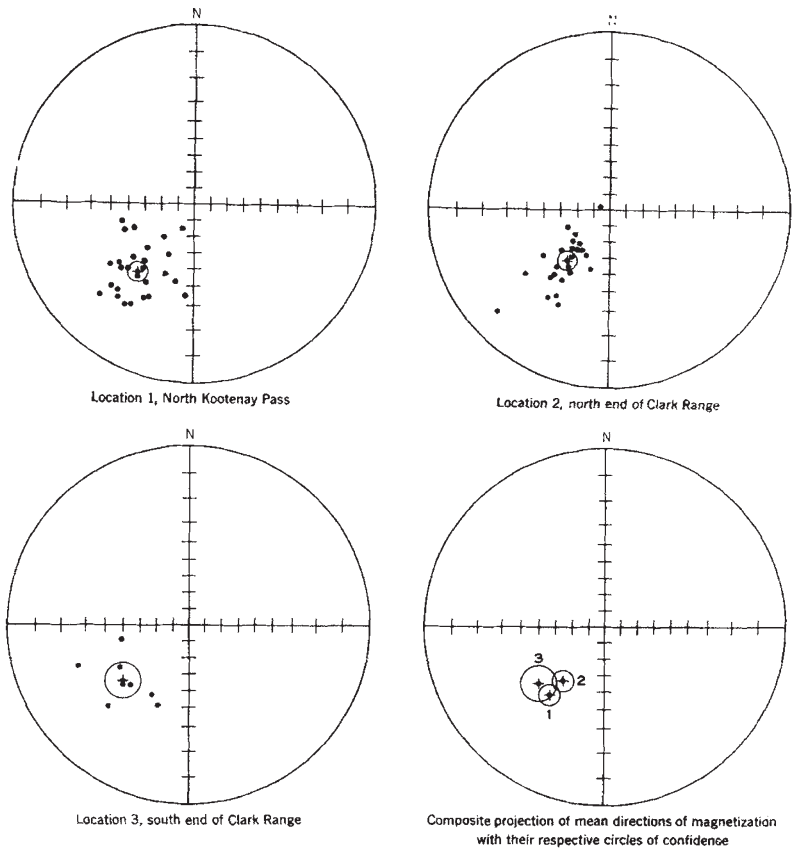


Fig. 2. Stereographic projections of the directions of magnetization for the three locations and composite projection of their respective mean directions and circles of confidence. ●, vectors pointing downwards; +, mean direction of magnetization

All cubes were subjected to decreasing-amplitude-alternating-field demagnetization and the maximum amplitude of the field was progressively increased after each set of measurements until no significant change (less than 5°) occurred in the magnetization direction in the cubes. This approximately stable state of magnetization was reached at 156 œrsteds.

Results from this demagnetization treatment indicate that the Kintla formation, where sampled, is magnetically stable, as the directions of magnetization for the majority of cubes were not significantly altered. After demagnetization the directions in the remainder became consistent with those on which demagnetization had little or no effect.

The mean palæomagnetic results for the three localities sampled are indicated in Table 1 and plotted in Fig. 2. Solid circles plotted on the Wulff stereographic projections in Fig. 2 represent the directions of the magnetic vectors with their positive pole pointing downwards. The mean direction for each group of samples is represented by a solid circle centred by a cross. The statistical parameters computed for the three groups of measurements are N, θ, and K as defined by Fisher[2], where N is the number of samples, θ is the radius of the circle of confidence calculated at the 95 per cent level of confidence (represented in Fig. 2 as open circles on the lower hemisphere of the stereographic projection), and K is a measure of the dispersion. All measurements are with respect to bedding, which was assumed to be horizontal at the time of deposition.

The fact that the calculated pole positions included in Table 1 agree well with those pole positions determined from Purcell and other Late Precambrian North American rocks[3,4] would suggest that the remanent magnetization has been preserved since Precambrian times. The results may therefore be applied with reasonable assurance to test for the possibility of differential rotation in the Lewis thrust plate in the early tertiary.

The directions of the magnetic vectors of samples from member *A* were in all cases found to point upwards and have been adjusted (sign of the inclination changed from negative to positive and 180° added

Table 1. Mean Palæomagnetic Results for Individual Collecting Sites

Collecting site	Location 1 North Kootenay Pass	Location 2 North end of Clark Range	Location 3 South end of Clark Range
Formation and member	Kintla *A*	Kintla *A* and *C*	Kintla *A*
Polarity*	−	− and +	−
Number of samples	27	24	8
Mean declination	218·5°	218·0°	230·0°
Mean inclination	+39·5°	+49·0°	+38·0°
θ	5·5°	6·5°	9·5°
K	23	24	40
Ancient pole position Lat.	10·5° S.	3·0° S.	6·5° S.
Ancient pole position Long.	150·5° W.	146·5° W.	159·0° W.

* For comparative purposes all vectors have been adjusted to point downward.

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to the declination) to point downward for comparison of results from members *A* and *C*.

It is evident from the composite projection in Fig. 2 that the circle of confidence for any one group of observations is intersected by those of the other two groups. This would suggest that within the limits of error of collection and measurement, which is between 5° and 10°, little if any relative rotation of the Kintla formation has occurred between any of the three collection sites. The differences in the mean declination angles are negligible between Flathead Range and the north end of the Clark Range and has a maximum of 12° between the latter and the south end of the Clark Range. It should be recalled that if the change in trend of the surface trace of the Lewis thrust were due entirely to differential rotation a disparity in the declination of the order of 35° should be indicated.

In summary, the continuity of surface structural and stratigraphic units throughout the length of the Lewis thrust plate in the area under investigation would suggest that the plate moved as a single, coherent, tectonic unit with no major transcurrent or transverse normal faulting. Palæomagnetic data, moreover, support this contention as there was, within the limits of error, no measurable differential rotation of the plate between the Flathead and the Clark and Lewis Ranges. It appears, therefore, that the Clark and Lewis Ranges were always structurally farther east than the Flathead Range. The present configuration of the eastern limit of the Lewis thrust plate is apparently genetically related to the geometry of the thrust surface established during the initial stages of movement rather than to differential erosion along the eastern margin of the plate.

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LOSS OF FLUORESCENT PARTICLES IN ATMOSPHERIC DIFFUSION EXPERIMENTS BY COMPARISON WITH RADIOXENON TRACER

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ZINC cadmium sulphide, disseminated in a finely ground particulate form, mainly in the size-range 1–5μ, has been widely used as a tracer in work on diffusion and atmospheric pollution. This fluorescent tracer technique was described independently by Perkins *et al.*[1] and Braham *et al.*[2] in 1952. The various problems in which it has been used can be conveniently divided into two groups. The first includes investigations of the spread of airborne substances released into the atmosphere. Here, only relative measurements of the concentrations of the particles are important in a particular trial, and provided any comparison of these is made after equivalent times of travel, then any time-decay of the tracer in a physical or chemical sense is unlikely to introduce error. The second is of particular importance in air-pollution investigations and includes all problems where the tracer is used to estimate the absolute concentration at any position resulting from a release of known size. Here, the decay characteristics of the tracer must be well defined.

Earlier workers[1,3] considered that zinc cadmium sulphide was stable in the presence of normally occurring concentrations of atmospheric pollutants for at least the first few hours after release. However, some work by one of the present authors (N. T.), to be reported elsewhere, in which vertical profiles of the particles were investigated some tens of kilometres down wind of releases on a long cross-wind line, showed an apparent decrease in the number of airborne particles with increasing distances of travel. A more reliable estimate of the loss was required, and it was decided to compare the fluorescent tracer directly with an inert gaseous tracer. Radioactive xenon-133 (half-life 5·27 days) was selected because of ease of sampling and detection.

The xenon and zinc cadmium sulphide tracers were released simultaneously at a uniform rate and sampled simultaneously at two distances down-wind. The total amount (number of particles or activity) of either tracer collected, divided by the sampling rate, is a measure of the dosage (time-integral of volumetric concentration) of that tracer at the point in question. If D_1, D_2 are the dosages of xenon at the near and far sampling distances respectively (units, μμc. min./m.^{3}) and d_1, d_2 the corresponding dosages of particles (units, particle min./m.^{3}) then L, the percentage loss of particles, is given by:

$$L = \left(1 - \frac{D_1}{D_2} \Big/ \frac{d_1}{d_2}\right) \times 100$$

This result is independent of the position of either sampling station with respect to the plume axis.

On each of three days, approximately 10 kgm. of zinc cadmium sulphide (Derby Luminescents 1318/10) and 3 curies of xenon-133 were emitted from a 20-metre incinerator stack at the Atomic Energy Research Establishment, Harwell, over a period of about 2 hours beginning near noon. Calculation showed that the health hazard to a person situated at the point of maximum ground-level concentration was negligible. During the three trials reported here, the wind directions varied between south-west and west-north-west with mean speeds of between 4 and 7 m./sec. Vigorous convection extended to heights of up to 1,700 m. and there was about one-half cloud cover present.

Sampling of both tracers took place at two stations placed one about 16 km. and the other about 60 km.

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